

MALMESBURY, South AfricaWMO: **687150**Lat: **33.472S**Lon: **18.718E**Elev: **101**StdP: **100.12**Time Zone: **2.00 (E02)**Period: **98-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.4	3.6	0.3	3.9	10.6	1.3	4.2	9.7	6.2	14.1	5.4	15.0	0.8	30	0.353

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.2	35.5	19.7	33.4	19.2	31.5	18.7	21.3	31.3	20.5	29.9	19.9	28.7	2.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.4	13.4	23.3	17.6	12.8	23.1	17.0	12.2	22.7	62.1	31.3	59.4	29.9	57.4	28.8	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5.9	5.0	4.3	1.0	40.5	0.8	1.7	0.4	41.7	0.0	42.7	-0.5	43.6	-1.0	44.9
			0.7	22.4	0.9	0.7	0.0	22.9	-0.5	23.4	-1.0	23.8	-1.7	24.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.6	23.5	23.8	21.7	18.8	15.5	12.5	11.9	11.9	14.0	17.3	19.5	21.9
	DBStd	5.14	3.03	3.15	2.98	3.46	2.37	2.14	2.33	2.00	2.74	3.45	3.07	2.86
	HDD10.0	17	0	0	0	0	0	3	7	5	1	0	0	0
	HDD18.3	934	1	1	5	34	94	176	199	201	134	62	23	4
	CDD10.0	2808	418	385	363	264	170	78	67	63	121	226	285	369
	CDD18.3	683	161	153	109	49	6	0	1	0	4	29	58	114
	CDH23.3	5728	1417	1257	727	381	44	4	8	4	47	323	508	1007
CDH26.7	2325	640	567	262	140	6	0	1	0	12	118	184	396	
Wind	WSAvg	1.6	2.0	1.8	1.6	1.3	1.5	1.5	1.4	1.6	1.5	1.6	1.9	1.9
Precipitation	PrecAvg	463	11	10	19	41	60	82	78	64	44	28	17	15
	PrecMax	810	48	44	70	134	156	211	205	236	110	88	103	64
	PrecMin	257	0	0	0	0	7	7	26	16	8	0	0	0
	PrecStd	116	10	12	19	32	41	48	43	39	25	25	22	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.9	37.8	35.4	34.6	27.3	23.6	24.1	23.5	28.8	33.8	34.8	36.3
		MCWB	21.0	19.9	19.7	18.9	15.9	14.8	14.7	14.9	16.9	18.1	18.2	19.6
	2%	DB	35.2	34.7	32.1	30.4	24.1	20.4	20.7	19.7	24.1	30.0	31.2	33.1
		MCWB	20.3	19.7	19.0	17.3	15.3	14.0	14.0	14.0	14.8	17.0	17.2	18.7
	5%	DB	32.7	32.3	29.6	27.2	21.8	18.1	18.2	17.8	21.3	27.0	28.6	30.8
		MCWB	19.6	19.4	18.3	16.8	15.0	13.1	13.1	13.0	14.2	16.4	16.9	18.3
	10%	DB	30.2	30.1	27.3	24.8	19.8	16.4	16.3	16.1	19.1	24.1	26.1	28.7
		MCWB	19.2	19.1	17.8	16.2	14.5	12.6	12.3	12.1	13.6	15.4	16.5	18.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.3	22.1	20.7	19.6	17.6	15.9	16.2	16.1	17.8	19.5	19.7	21.1
		MCDB	33.1	32.5	30.9	29.9	23.5	20.5	21.9	21.3	25.1	30.1	29.5	31.1
	2%	WB	21.5	21.1	20.0	18.7	16.5	14.7	14.6	14.7	16.0	18.3	18.7	20.2
		MCDB	32.1	30.6	29.7	28.0	21.1	18.3	19.4	18.8	22.8	27.8	27.5	29.6
	5%	WB	20.7	20.3	19.2	17.7	15.8	14.0	13.6	13.5	14.9	17.0	18.0	19.6
		MCDB	30.1	29.0	27.1	24.7	19.5	16.8	17.2	16.9	20.0	25.0	26.1	28.3
	10%	WB	19.9	19.7	18.5	17.0	15.1	13.4	12.8	12.6	14.0	16.0	17.2	18.8
		MCDB	28.3	27.8	25.6	23.0	18.6	15.8	15.6	15.4	18.4	22.4	24.3	26.6
Mean Daily Temperature Range	5% DB	MDBR	13.2	13.2	12.5	12.2	9.9	9.9	10.6	9.8	10.9	12.7	12.7	13.1
		MCDBR	17.5	16.7	15.5	16.4	13.8	13.2	14.5	13.6	15.1	17.8	17.7	17.1
		MCWBR	5.2	4.7	4.9	5.8	6.7	7.7	8.7	8.3	7.2	6.8	6.2	5.6
	5% WB	MCDBR	14.9	13.8	13.8	14.2	11.0	10.4	12.4	11.9	13.0	15.7	15.2	14.6
		MCWBR	4.7	4.5	4.9	5.6	6.0	6.5	7.9	7.6	6.8	6.5	5.7	5.3
Clear-Sky Solar Irradiance	taub	0.332	0.333	0.328	0.315	0.311	0.305	0.307	0.326	0.335	0.344	0.329	0.333	
	taud	2.509	2.510	2.520	2.557	2.562	2.553	2.538	2.483	2.425	2.414	2.498	2.500	
	Ebn at Noon	1005	984	954	912	861	839	856	885	930	961	1002	1008	
	Edh at Noon	113	109	101	86	76	72	77	91	109	119	114	115	
All-Sky Solar Radiation	RadAvg	8.37	7.52	6.09	4.43	3.08	2.60	2.90	3.63	5.00	6.58	7.81	8.43	
	RadStd	0.20	0.20	0.20	0.23	0.20	0.15	0.18	0.17	0.20	0.33	0.22	0.23	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)		+0.36	+0.49	N/A	N/A	+0.53	+0.47	-2	-74	+111	+61

Nomenclature: See separate page